

Google Colab Wonders

Tips, Tricks & Powerful Integrations!

Supercharge your Colab notebooks with these amazing features

Forms & Inputs · Camera Capture · JavaScript Power · Drive & Files · Cloud & APIs · Visualizations · Web Apps

1. FORM FIELDS (USER INPUT)

String: example
Slider: 142
Number: 102
Date: 2010-11-05
Dropdown: thursday
Select/Input: oranges

Forms support many types of fields.

2. CAMERA CAPTURE



Capture

Capture photos using your webcam directly in Colab!

Saved to photo.jpg

3. CROSS-OUTPUT COMMUNICATION

Listener (receives)

```
const listenerChannel =
new BroadcastChannel('channel');
listenerChannel.onmessage =
 (msg) => {
  document.body.appendChild(
   Object.assign(document.createElement('div'),
    {textContent: msg.data})
  );
};
```

Sender (sends)

```
const senderChannel =
new BroadcastChannel('channel');

senderChannel.postMessage(
 'Hello world!'
);
```

Output: Hello world!



4. display.JAVASCRIPT

Inject JavaScript to interact with the notebook output.

```
import IPython
js_code = '''
document.querySelector("#output-area")
 .appendChild(document.createTextNode("hello world!"));
'''

display(IPython.display.Javascript(js_code))
```

Output: Hello world!

5. GOOGLE DRIVE & FILES

Download from Drive

```
drive.CreateFile({'id': file_id})
 .GetContentFile('myfile.txt')
```

Upload local file

```
from google.colab import files
files.upload()
```

Mount Drive

```
from google.colab import drive
drive.mount('/gdrive')
```

6. EVALUATE JS EXPRESSIONS

Run JavaScript and get the result in Python.

```
display(Javascript("""
 window.someValue = new Promise(resolve => {
  setTimeout(() => resolve("hello world!"), 100);
 });
"""))

from google.colab import output
value = output.eval_js('someValue')
```

Output: hello world!

7. HIDE CODE

Hide cells using display-mode: "form".

```
#@title Give me a name {display-mode: "form"}

# This code will be hidden
a, b = 7, 6
```

8. IMPORT MISSING LIBRARIES

Install system & Python packages easily.

```
!pip install -q matplotlib-venn
!apt-get -qq install -y libfluidsynth1
```

9. GOOGLE SHEETS → PANDAS

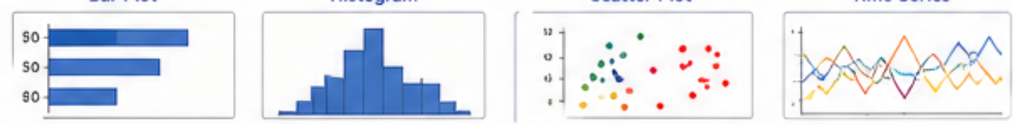
→ pandas

- Authenticate & open sheet
- Read all values
- Convert to DataFrame

Date	Project A	Project B	Totals
Mon	2	3	5
Tue	1	4	5
Total hrs	3	7	10

10. DATA VISUALIZATIONS (ALTair)

Bar Plot · Histogram · Scatter Plot · Time Series



Linked Views



Stacked Histogram



Easy & interactive charts with Altair!

11. ASYNCIO & CONCURRENCY

Asyncio (I/O-bound)

```
async with aiohttp.
 ClientSession() as s:
  async with s.get(url)
   as r:
    await r.text()
```

ThreadPoolExecutor (Blocking I/O)

```
loop.run_in_executor(
  ThreadPoolExecutor(),
  blocking_task, arg)
```

ProcessPoolExecutor (CPU-bound)

```
loop.run_in_executor(
  ProcessPoolExecutor(),
  cpu_task,  arg)
```

CPU

✓ Run tasks concurrently for better performance!

12. FILE DOWNLOADS

Download from URL / GDrive / GCS

```
!gdown --id FILE_ID -O name.h5

!gsutil cp gs://bucket/file.txt ./
```

Download to local file

```
from google.colab import files
files.download('file.txt')
```

13. WEBSERVERS & APIS (FLASK + NGROK)

Simple Web App

```
from flask import Flask
app = Flask(__name__)

@app.route('/')
def home():
   return "<h1>Hello Colab!</h1>"
app.run()
```

Run with NGROK → Public URL!

ngrok →

REST API Example

```
GET /todos
POST /todos
PUT /todos/<id>
DELETE /todos/<id>
```

Run:

```
!python app.py &
npx localtunnel --port 5000 \
  --subdomain romellfudi
```

14. JS ↔ PYTHON COMMUNICATION

Python Callback

```
output.register_callback(
 'notebook.Concat', Concat)
```

JavaScript Call

```
google.colab.kernel.
invokeFunction(
  'notebook.Concat',
  ['hello', 'world!"], {}
)
```

JS

Result: hello world!

15. LIST FILES FROM GOOGLE DRIVE

```
listed = drive.ListFile({'q': "title
contains '.json'"}).GetList()
for file in listed:
   print(file['title'])
```

16. CLOUD & API INTEGRATIONS

BigQuery
- Query with SQL
- Read to Pandas

Cloud Storage
- Download files
- Upload & manage

17. GITHUB INTEGRATION

```
!git clone -l -s
git://github.com/user/repo.git
!curl -L -O
https://raw.githubeurcotment.com/.../file.csv
```

18. UTILITY TOOLS

- 7-Zip (libarchive)
- GraphViz & PyDot
- CV2 Image Display
- TensorFlow / Venn / More!

19. SERVER INNER (ADVANCED)

Run a small HTTP server inside Colab and serve custom content.

20. HANDLER OUTPUT PRINT

```
Starting...
working...
still working...
All done!
```